Screen Flow Map
Bosque La Primavera · Respuesta y Monitoreo

Stage 2 · Mobile Development · 20 screens, 3 roles, 6 back-end sources
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ARROWS

- Navigation (user action)
- System-triggered (push / redirect)
- Error / restricted path
- Number = action in the screen card
- Back / tab-bar navigation

BACK-END SOURCES

- AUTH Authentication & Authorization (OAuth 2.0 · RBAC)
- API REST / GraphQL API layer (business logic, CRUD, filters)
- GEO Geospatial database (PostGIS / MongoDB Geospatial)
- FILES File & asset storage (S3 / Firebase · map tiles, PDFs, photos)
- SYNC Sync & Queue engine (offline uploads, retries)
- DEVICE On-device (GPS sensor, local cache / queue)

A1 Inicio de sesión

Login · 01 Acceso

PURPOSE

Authenticate staff with their institutional account and route each user to the home of their role.

CONTENTS

- Institutional e-mail and password (show / hide)
- Remember session, recover access
- Demo-environment disclaimer

ACTIONS

1. Sign in as Field Responder → **R1 Inicio del rescatista**
2. Sign in as Coordinator → **C1 Resumen operativo**
3. Sign in as Administrator → **AD1 Inicio del administrador**
4. Wrong credentials / locked account → *inline error message*
5. Recuperar acceso → *reset e-mail sent*

BACK-END CALLS

- AUTH `POST /auth/token (OAuth 2.0 + PKCE) → JWT with role claim`
- AUTH `POST /auth/recover`
- DEVICE `Secure storage of refresh token`



R1 Inicio del rescatista

Responder home · 08 Perfiles

PURPOSE

Show the responder's active mission, quick actions and own position, even offline.

CONTENTS

- Duty status and shift
- Active mission card (INC-026-041, ETA 12 min)
- Quick actions: Reportar / Abrir mapa
- Mini-map with own GPS position
- Offline banner (3 pending changes)

ACTIONS

1. Ver misión y actualizar estado → **R4 Detalle de incidente**
2. Reportar → **R5 Registrar incidente**
3. Abrir mapa / Centrar mapa → **R2 Mapa operativo**
4. Tab · Incidentes → **R3 Lista de incidentes**
5. Tab · Perfil → **R7 Mi perfil**

BACK-END CALLS

- API `GET /me/mission (active assignment)`
- GEO `GET /incidents/{id}/location`
- FILES `GET /tiles/{z}/{x}/{y} (cached)`
- DEVICE `GPS fix + pending queue count`
- SYNC `GET /sync/status`



R2 Mapa operativo

Operational GIS map · 02 Operación

PURPOSE

Interactive GIS map of trails, access roads and georeferenced incident pins.

CONTENTS

- Search (trail, zone, incident)
- Base layers: forest, contour lines, streams, roads, trails
- Pins by type & severity (rescue, smoke, search)
- Zoom / layers / locate controls, legend
- Selected-incident card

ACTIONS

1. Tap pin / Ver detalle → **R4 Detalle de incidente**
2. Pin of another incident → **R3 Lista de incidentes**
3. Iniciar ruta al incidente · 1,8 km → *route drawn on map*
4. Search / layers / zoom → *map updates*
5. Tab · Inicio → **R1 Inicio del rescatista**

BACK-END CALLS

- FILES `GET /tiles/{z}/{x}/{y} (base map, offline packs)`
- GEO `GET /incidents?bbox=&status=active`
- GEO `GET /trails?q= (search)`
- API `GET /routes?from=&to= (trail routing)`
- DEVICE `GPS (my location)`



R4 Detalle de incidente

Incident detail · 02 Operación

PURPOSE

Full incident record: location, assigned team and timeline; update the mission status.

CONTENTS

- Severity & status tags
- Description, location, GPS precision
- Mini-map with incident and own position
- Assigned brigade with ETA
- Timeline of events

ACTIONS

1. Ver mapa → **R2 Mapa operativo**
2. Actualizar estado → *status changes (queued if offline)*
3. Añadir avance → *new timeline event*
4. Back (←) → **R3 Lista de incidentes**
5. Edit without permission (RBAC) → **E2 Acceso denegado**

BACK-END CALLS

- API `GET /incidents/{id}`
- API `GET /incidents/{id}/timeline`
- GEO `GET /brigades/{id}/position`
- API `PATCH /incidents/{id}/status · POST /incidents/{id}/events`
- SYNC `Enqueue changes when offline`



R3 Lista de incidentes

Incident list · 02 Operación

PURPOSE

Prioritized list of incidents to find and open a case quickly.

CONTENTS

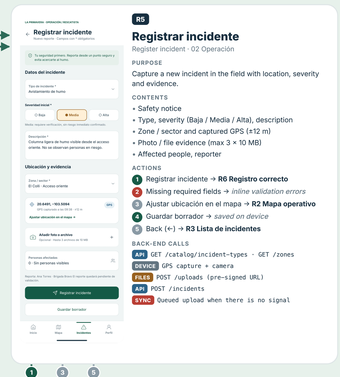
- Search by name, folio or zone
- Filter chips: Activos 12 / Misiones 1 / Cerrados 36
- Sort: priority and most recent
- Incident cards (severity, status, owner)

ACTIONS

1. Tap incident card → **R4 Detalle de incidente**
2. Registrar incidente → **R5 Registrar incidente**
3. Search / filters → *list re-queried*
4. Tab · Mapa → **R2 Mapa operativo**

BACK-END CALLS

- API `GET /incidents?status=&q=&sort=priority&page=`
- GEO `Zone names resolved from coordinates`
- DEVICE `Cached list for offline reading`



R5 Registrar incidente

Register incident · 02 Operación

PURPOSE

Capture a new incident in the field with location, severity and evidence.

CONTENTS

- Safety notice
- Type, severity (Baja / Media / Alta), description
- Zone / sector and captured GPS (±12 m)
- Photo / file evidence (max 3 × 10 MB)
- Affected people, reporter

ACTIONS

1. Registrar incidente → **R6 Registro correcto**
2. Missing required fields → *inline validation errors*
3. Ajustar ubicación en el mapa → **R2 Mapa operativo**
4. Guardar borrador → *saved on device*
5. Back (←) → **R3 Lista de incidentes**

BACK-END CALLS

- API `GET /catalog/incident-types · GET /zones`
- DEVICE `GPS capture + camera`
- FILES `POST /uploads (pre-signed URL)`
- API `POST /incidents`
- SYNC `Queued upload when there is no signal`



R6 Registro correcto

Registration success · 07 Estados

PURPOSE

Confirm that the incident was saved and is pending validation.

CONTENTS

- Success seal
- Folio INC-026-049 and summary
- Status: pending validation
- Offline note (part of 3 pending changes)

ACTIONS

1. Ver incidente registrado → **R3 Lista de incidentes**
2. Volver al mapa → **R2 Mapa operativo**

BACK-END CALLS

- API `Response of POST /incidents (folio, status)`
- SYNC `Notify coordination when synced`

FIELD RESPONDER
Paramedic volunteer · 20-35



R7 Mi perfil

User profile · 05 Cuenta

PURPOSE

Account information and device preferences for the responder.

CONTENTS

- Identity, role and brigade
- Toggles: mission notifications, share location, mobile data
- Sync & maps, accessibility, language, security
- Log out

ACTIONS

1. Sincronización y mapas → **R8 Sincronización**
2. Admin-only function (roles) → **E2 Acceso denegado**
3. Cerrar sesión → **A1 Inicio de sesión**
4. Toggle preferences → *saved instantly*

BACK-END CALLS

- API `GET /me · PATCH /me/preferences`
- AUTH `POST /auth/logout (revoke tokens)`
- DEVICE `Accessibility / high-contrast mode`



R8 Sincronización

Offline sync · 04 Conectividad

PURPOSE

Show what is stored on the device and send pending changes when there is signal.

CONTENTS

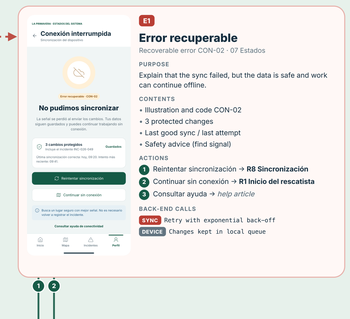
- Connection mode and last sync time
- Pending changes queue (3)
- Downloaded map packages
- Local storage usage (80 / 500 MB)

ACTIONS

1. Comprobar conexión y sincronizar (ok) → *queue emptied*
2. Sync fails / signal lost → **E1 Error recuperable**
3. Gestionar mapas → *download / delete packs*
4. Back (←) → **R7 Mi perfil**

BACK-END CALLS

- SYNC `POST /sync/batch (queued incidents, events, positions)`
- FILES `GET /tiles/packages (offline areas)`
- DEVICE `Local queue (IndexedDB / SQLite)`



E1 Error recuperable

Recoverable error CON-02 · 07 Estados

PURPOSE

Explain that the sync failed, but the data is safe and work can continue offline.

CONTENTS

- Illustration and code CON-02
- 3 protected changes
- Last good sync / last attempt
- Safety advice (find signal)

ACTIONS

1. Reintentar sincronización → **R8 Sincronización**
2. Continuar sin conexión → **R1 Inicio del rescatista**
3. Consultar ayuda → *help article*

BACK-END CALLS

- SYNC `Retry with exponential back-off`
- DEVICE `Changes kept in local queue`



E2 Acceso denegado

Access denied SEG-403 · 07 Estados

PURPOSE

Tell the user that a function needs another role, without losing the session.

CONTENTS

- Shield illustration, code SEG-403
- Current user and role
- What this role can do
- Guidance: request a role review

ACTIONS

1. Volver a mi inicio operativo → **R1 Inicio del rescatista**
2. Contactar a coordinación → *message to coordinator*

BACK-END CALLS

- AUTH `RBAC check returns HTTP 403`
- API `POST /access-requests (role review)`

C1 Resumen operativo

Coordinator dashboard · 03 Coordinación

PURPOSE

Operational summary of the period to decide where to send resources.

CONTENTS

- KPIs: active incidents, brigades in field, reports, avg. response
- Incidents per interval (bar chart)
- Response status (donut)
- Available brigades

ACTIONS

1. Comparar → **C2 Comparación histórica**
2. Asignar → (available brigades) → **C4 Brigadas y misiones**
3. Tab · Zonas → **C3 Zonas críticas**
4. Tab · Informes → **C5 Reportes**
5. Too little data → *data-confidence alert*

BACK-END CALLS

- API `GET /stats/summary?from=&to=`
- GEO `Aggregations by zone // interval`
- API `GET /brigades?status=available`



C2 Comparación histórica

Period comparison · 03 Coordinación

PURPOSE

Compare equivalent periods to detect trends and critical hotspots.

CONTENTS

- Period chips: monthly / quarterly / custom
- Current vs. reference period
- Paired bar chart (volume)
- Indicator table and insight

ACTIONS

1. Change period → *charts recalculated (< 5 s)*
2. Insufficient data → *data-confidence alert*
3. Back (←) → **C1 Resumen operativo**

BACK-END CALLS

- API `GET /stats/compare?current=&reference=`
- GEO `Spatio-temporal aggregation queries`



C3 Zonas críticas

Critical zones · 03 Coordinación

PURPOSE

See where incidents concentrate and which zones have priority.

CONTENTS

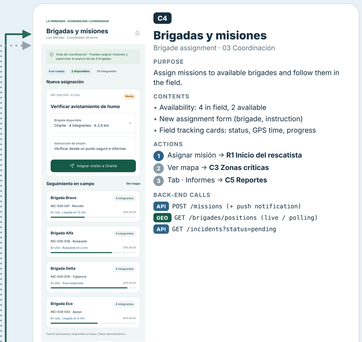
- Filters: severity / concentration / brigades
- Map with severity polygons and markers
- Severity legend
- Priority list of zones A–C

ACTIONS

1. Tap marker / zone → **R4 Detalle de incidente**
2. Filters, zoom, layers → *map updates*
3. Tab · Brigadas → **C4 Brigadas y misiones**

BACK-END CALLS

- GEO `GET /zones/hotspots?metric=severity`
- GEO `GET /incidents?bbox=`
- FILES `GET /tiles/{z}/{x}/{y}`



C4 Brigadas y misiones

Brigade assignment · 03 Coordinación

PURPOSE

Assign missions to available brigades and follow them in the field.

CONTENTS

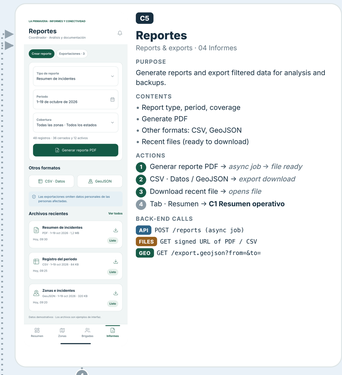
- Availability: 4 in field, 2 available
- New assignment form (brigade, instruction)
- Field tracking cards: status, GPS time, progress

ACTIONS

1. Asignar misión → **R1 Inicio del rescatista**
2. Ver mapa → **C3 Zonas críticas**
3. Tab · Informes → **C5 Reportes**

BACK-END CALLS

- API `POST /missions (+ push notification)`
- GEO `GET /brigades/positions (live / polling)`
- API `GET /incidents?status=pending`



C5 Reportes

Reports & exports · 04 Informes

PURPOSE

Generate reports and export filtered data for analysis and backups.

CONTENTS

- Report type, period, coverage
- Generate PDF
- Other formats: CSV, GeoJSON
- Recent files (ready to download)

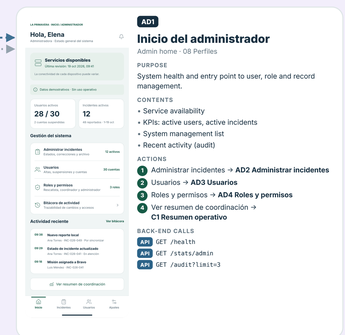
ACTIONS

1. Generar reporte PDF → *async job → file ready*
2. CSV · Datos / GeoJSON → *export download*
3. Download recent file → *opens file*
4. Tab · Resumen → **C1 Resumen operativo**

BACK-END CALLS

- API `POST /reports (async job)`
- FILES `GET signed URL of PDF / CSV`
- GEO `GET /export.geojson?from=&to=`

COORDINATOR
Dispatch manager · 35–40



AD1 Inicio del administrador

Admin home · 08 Perfiles

PURPOSE

System health and entry point to user, role and record management.

CONTENTS

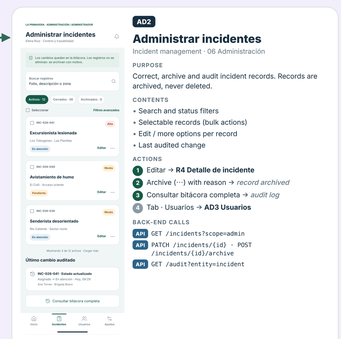
- Service availability
- KPIs: active users, active incidents
- System management list
- Recent activity (audit)

ACTIONS

1. Administrar incidentes → **AD2 Administrar incidentes**
2. Usuarios → **AD3 Usuarios**
3. Roles y permisos → **AD4 Roles y permisos**
4. Ver resumen de coordinación → **C1 Resumen operativo**

BACK-END CALLS

- API `GET /health`
- API `GET /stats/admin`
- API `GET /audit?limit=3`



AD2 Administrar incidentes

Incident management · 06 Administración

PURPOSE

Correct, archive and audit incident records. Records are archived, never deleted.

CONTENTS

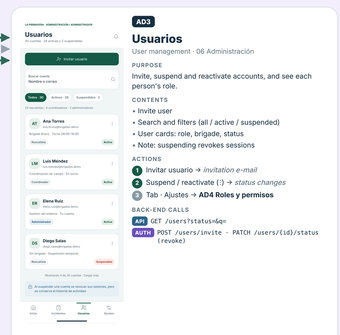
- Search and status filters
- Selectable records (bulk actions)
- Edit / more options per record
- Last audited change

ACTIONS

1. Editar → **R4 Detalle de incidente**
2. Archive (···) with reason → *record archived*
3. Consultar bitácora completa → *audit log*
4. Tab · Usuarios → **AD3 Usuarios**

BACK-END CALLS

- API `GET /incidents?scope=admin`
- API `PATCH /incidents/{id} · POST /incidents/{id}/archive`
- API `GET /audit?entity=incident`



AD3 Usuarios

User management · 06 Administración

PURPOSE

Invite, suspend and reactivate accounts, and see each person's role.

CONTENTS

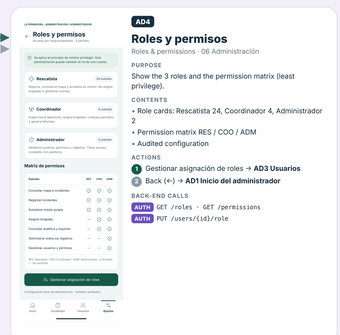
- Invite user
- Search and filters (all / active / suspended)
- User cards: role, brigade, status
- Note: suspending revokes sessions

ACTIONS

1. Invitar usuario → *invitation e-mail*
2. Suspend / reactivate (⋮) → *status changes*
3. Tab · Ajustes → **AD4 Roles y permisos**

BACK-END CALLS

- API `GET /users?status=&q=`
- AUTH `POST /users/invite · PATCH /users/{id}/status (revoke)`



AD4 Roles y permisos

Roles & permissions · 06 Administración

PURPOSE

Show the 3 roles and the permission matrix (least privilege).

CONTENTS

- Role cards: Rescatista 24, Coordinador 4, Administrador 2
- Permission matrix RES / COO / ADM
- Audited configuration

ACTIONS

1. Gestionar asignación de roles → **AD3 Usuarios**
2. Back (←) → **AD1 Inicio del administrador**

BACK-END CALLS

- AUTH `GET /roles · GET /permissions`
- AUTH `PUT /users/{id}/role`

ADMINISTRATOR
Data & admin operator · 22–40

Where the data comes from (back-end services · Stage 1)

SOURCE	ROLE IN THE SYSTEM	SCREENS THAT CALL IT
AUTH	Authentication & Authorization (OAuth 2.0 · RBAC)	A1 · R7 · E2 · AD3 · AD4
API	REST / GraphQL API layer (business logic, CRUD, filters)	R1 · R2 · R4 · R3 · R5 · R6 · R7 · E2 · C1 · C2 · C4 · C5 · AD1 · AD2 · AD3
GEO	Geospatial database (PostGIS / MongoDB Geospatial)	R1 · R2 · R4 · R3 · C1 · C2 · C3 · C4 · C5
FILES	File & asset storage (S3 / Firebase · map tiles, PDFs, photos)	R1 · R2 · R5 · R8 · C3 · C5
SYNC	Sync & Queue engine (offline uploads, retries)	R1 · R4 · R5 · R6 · R8 · E1
DEVICE	On-device (GPS sensor, local cache / queue)	A1 · R1 · R2 · R3 · R5 · R7 · R8 · E1

Notes

Offline-first. Field screens (R1–R8) read from the local cache and write to a local queue. The Sync & Queue engine sends the queue when there is signal; if it fails, the user goes to E1 and nothing is lost.

RBAC. Every API call carries the JWT from A1. If the role lacks a permission, the API answers 403 and the app shows E2 (SEG-403).

Data confidence. C1/C2 show an alert when the selected period has too few records for a reliable comparison (Stage 1 NFR).

Performance targets. Spatial queries and period comparisons < 5 s; up to 50 concurrent field operators and dispatchers.